Dictionary

Facility: Represents a railway facility such as a station, depot, or yard.

Attribute	Type	Description
ID (PK)	number(38)	Unique identifier of the facility.
Name	varchar2(255)	Name of the facility (e.g., “São Romão”).
TypeID (FK)	number(10)	References **FacilityType**. Indicates the facility’s type (station, depot, etc.).
Address	varchar2(255)	Physical address of the facility.
City	varchar2(255)	City where the facility is located.
Capacity	number(10)	Maximum capacity (e.g., trains, wagons, or passengers).
Electrified	char(3)	Indicates if the facility is electrified (‘YES’/‘NO’).
OwnerVAT (FK)	varchar2(15)	References VAT number of the owner responsible for managing or maintaining the facility.

FacilityType: Defines the category of a facility (e.g., station, depot, terminal).

Attribute	Data Type	Description
FacilityTypeID (PK)	number(10)	Unique identifier for each facility type.
Name	varchar2(255)	Name of the type (e.g., “Station”, “Depot”).

Line: Represents a railway line connecting two facilities.

Attribute	Type	Description
ID (PK)	number(38)	Unique identifier for the line.
Name	varchar2(255)	Line name or designation.
OwnerVAT (FK)	varchar2(15)	References **Owner** responsible for the line’s maintenance and operation.
StartID (FK)	number(38)	References **Facility** where the line begins.
EndID (FK)	number(38)	References **Facility** where the line ends.
GaugeName (FK)	varchar2(255)	References Gauge entity.

Line_segment: Defines a segment or section of a railway line.

Attribute	Type	Description
ID (PK)	number(10)	Unique identifier for the segment.
LineID (FK)	number(38)	References **Line** that the segment belongs to.
Order	number(10)	The segment’s order in the line (1st, 2nd, etc.).
Electrified	char(3)	Indicates if the segment is electrified (‘YES’/‘NO’).
Max_weight	float(38)	Maximum allowed train weight on this segment (kg/m).
Length	number(38)	Segment length (m).
NumberOfTracks	number(10)	Number of parallel tracks in the segment.
SpeedLimit	number(10)	Maximum allowed speed on the segment.

Owner: Represents an organization that owns facilities or lines.

Attribute	Type	Description
VATnumber (PK)	varchar2(15)	VAT number of the company.
Name	varchar2(255)	Full name of the owning company.
ShortName	varchar2(255)	Abbreviated company name.

Operator: Represents a person or entity responsible for operating locomotives or wagons.

Attribute	Type	Description
VATnumber (PK)	varchar2(15)	VAT number of the operator.
Name (PK)	varchar2(255)	Full name of the operator company.
ShortName	varchar2(255)	Abbreviated operator name.

Gauge: Represents the track width standard used by locomotives, wagons, lines, or facilities.

Attribute	Type	Description
Name (PK)	varchar2(255)	Unique name of the gauge.
Gauge	number(10)	Width of the track in millimeters (e.g., 1435 for standard).

Wagon: Represents a railway wagon or freight car.

Attribute	Type	Description
WagonNumber (PK)	number(38)	Unique identifier for each wagon.
ModelID (FK)	number(100)	References **WagonModel** that defines its characteristics.
ServiceYear	number(4)	Year the wagon entered service.
OperatorVAT (FK)	varchar2(15)	References VAT number of the operator managing the wagon.

WagonModel: Defines the technical specifications of wagon models.

Attribute	Type	Description
ModelIDNumber (PK)	number(38)	Unique identifier for the wagon model.
Model	varchar2(255)	Model name or code.
Maker	varchar2(255)	Manufacturer of the wagon.
NumberBogies	number(1)	Number of bogies (wheel assemblies).
Bogies	varchar2(255)	Description or type of bogie configuration.
Length	number(10)	Wagon length (m).
Width	float(10)	Wagon width (m).
Height	float(10)	Wagon height (m).
Weight	float(10)	Total empty wagon weight (N).
MaxSpeed	number(10)	Maximum allowed speed.
Payload	float(10)	Maximum load capacity.
Volume	float(10)	Internal volume (m^3).
TypeID (FK)	number(38)	References **WagonType** to define wagon category.
GaugeName (FK)	varchar2(255)	References Gauge entity.

WagonType: Defines categories of wagons.

Attribute	Type	Description
ID (PK)	number(38)	Unique identifier for wagon type.
Name	varchar2(255)	Name of the type (e.g., Cereal wagon, Container wagon).

Locomotive: Represents an individual locomotive in service.

Attribute	Type	Description
LocomotiveNumber (PK)	number(38)	Unique identifier (fleet number) for each locomotive.
Name	varchar2(255)	Locomotive name or designation.
ModelID (FK)	number(38)	References **LocomotiveModel** that defines its specifications.
Service	number(4)	Service year or operational year.
OperatorVAT (FK)	varchar2(15)	References VAT identification number of the operator managing the locomotive.

LocomotiveModel: Defines a locomotive model's technical characteristics.

Attribute	Type	Description
ID (PK)	number(38)	Unique identifier for the locomotive model.
Make	varchar2(255)	Manufacturer of the locomotive
Model	varchar2(255)	Model name or number.
NumberBogies	number(10)	Number of bogies (wheel assemblies).
Bogies	varchar2(255)	Description of bogie configuration.
Power	number(10)	Engine power rating (kW).
Length	float(10)	Locomotive length (m).
Width	float(10)	Locomotive width (m).
Height	float(10)	Locomotive height (m).
Weight	number(10)	Total locomotive weight (N).
MaxSpeed	number(10)	Maximum operational speed.
OperationalSpeed	float(10)	Regular service speed.
Traction	number(10)	Force of traction (N).
DieselTypeID (FK)	number(38)	References **Diesel_locomotive** if diesel-powered.
ElectricTypeID (FK)	number(38)	References **Electric_locomotive** if electric-powered.
GaugeName (FK)	number(10)	References Gauge entity.

Diesel_locomotive: Represents specific data for diesel locomotive models.

Attribute	Type	Description
DieselTypeID (PK)	number(38)	Unique identifier for the diesel locomotive type.
Fuel_capacity	number(38)	Fuel tank capacity, combustivel (l).

Electric_locomotive: Represents specific data for electric locomotive models.

Attribute	Type	Description
ElectricTypeID (PK)	number(38)	Unique identifier for the electric locomotive type.
Voltage	number(10)	Operating voltage (V).
Frequency	number(10)	Frequency (Hz).

Glossary

General Concepts:

Term	Definition / Explanation
DBMS (Database Management System)	A software system that manages the creation, organization, and manipulation of a database. Examples: MySQL, PostgreSQL, Oracle.
Entity	A real-world object or concept represented as a table in the database (e.g., Facility, Line, Wagon).
Attribute	A property or characteristic of an entity (e.g., Name, ID, Capacity). Each attribute corresponds to a column in a table.
Primary Key (PK)	A unique identifier for each record in a table (e.g., ID, LocomotiveNumber).
Foreign Key (FK)	An attribute that creates a relationship between two entities by referencing another entity's primary key.
Relationship	A connection between entities (e.g., each Line connects two Facilities).
Cardinality	Defines how many instances of one entity relate to another (e.g., one Line can have many Segments).
Normalization	The process of structuring a database to reduce redundancy and improve data integrity.
VARCHAR(n)	Variable-length character data type; stores text up to *n* characters.
NUMBER(n)	Numeric data type; stores numeric values up to *n* digits.
FLOAT(n)	Floating-point data type; stores decimal numbers with precision.
CHAR(3)	Fixed-length character field (used for short codes like "Yes/No" or "Y/N").
U (Unique)	Constraint ensuring that all values in a column are distinct.
N (Nullable)	Indicates that the field can be left empty (NULL).

Relationships:

Relationship	Description
Facility – FacilityType	Each facility has one type. A type can be shared by multiple facilities.
Facility – Owner	Each facility is owned by one Owner.
Line – Facility	Each line connects two facilities (start and end).
Line – Segment	A line consists of one or many segments.
Line – Owner	A line has one owner responsible for maintenance.
Line – Gauge	Each line has one track gauge.
Wagon – WagonModel	Each wagon has one model type.
Locomotive – LocomotiveModel	Each locomotive has one model type.
LocomotiveModel – Diesel_locomotive / Electric_locomotive	A locomotive model can be either diesel or electric, not both.
LocomotiveModel – Gauge	Each locomotive model is designed for a specific gauge.
WagonModel – WagonType	Each wagon model belongs to one wagon type.
Operator – Locomotive / Wagon	An operator manages one or more locomotives and wagons.
WagonModel – Gauge	Each wagon model is designed for a specific gauge.

Acronyms:

Acronym	Meaning
PK	Primary Key
FK	Foreign Key
ID	Identifier
VAT	Value Added Tax (used in VAT Number for business identification)
DBMS	Database Management System
Y/N	Yes or No (used for attributes like Electrified)